MOS Capacitor

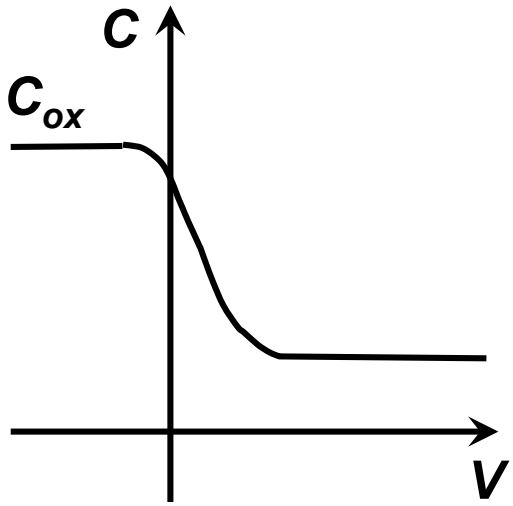
- Introduction
- Ideal MOS Capacitors
  - Qualitative Description of MOS Capacitor
  - Quantitative Analysis of MOS Capacitor
- *C-V* Characteristics of MOS Capacitor
- Non Ideal MOS-Capacitor
  - Defect Charges in MOS Capacitors
- Applications of *C-V* Curves

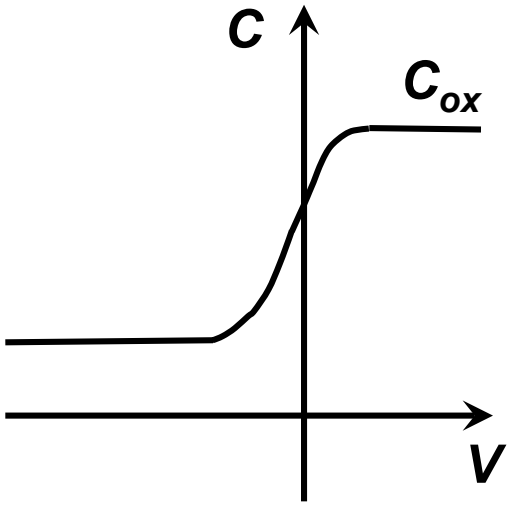
Applications of *C-V* Characteristics

- Extraction of Parameters of MOS Cap:
  - Substrate dopant type
  - Doping concentration
  - Flat band voltage and oxide thickness
  - Minority carrier lifetime
- Extraction of Oxide/Substrate Interface:
  - Mobile charge Q_M
  - Fixed oxide Q_f
  - Oxide trapped charge Q_{ot}
  - Interface charge Q_{it}

Substrate Dopant Type

- Measurement of substrate dopant type in semiconductor, using high-frequency *C-V.*



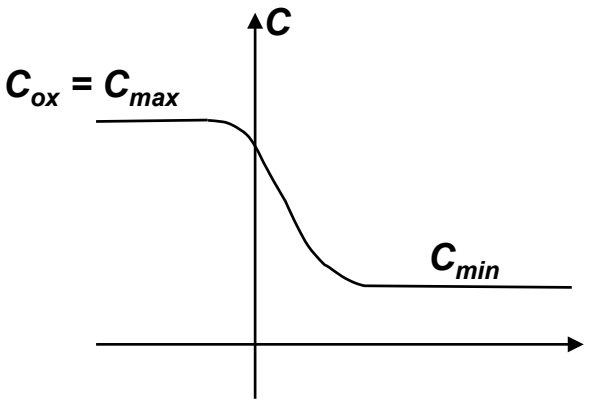


Substrate Doping Concentration

$$\frac{1}{C_{\min}} = \frac{1}{C_{ox}} + \frac{1}{C_{si,\min}} = \frac{1}{C_{\max}} + \frac{1}{\frac{\varepsilon_{si}}{W_m}} \quad \Rightarrow \quad W_m = \varepsilon_{si}\left(\frac{1}{C_{\min}} - \frac{1}{C_{\max}}\right)$$

In addition, $$W_m = \sqrt{\frac{2\varepsilon_{si} \cdot 2\Phi_B}{qN_A}} = \sqrt{\frac{4\varepsilon_{si} kT \ln\frac{N_A}{n_i}}{q^2 N_A}}$$

$$\Rightarrow \quad \sqrt{\frac{4\varepsilon_{si} kT \ln\frac{N_A}{n_i}}{q^2 N_A}} = \varepsilon_{si}\left(\frac{1}{C_{\min}} - \frac{1}{C_{\max}}\right)$$



- Not closed form, but can be solved by iteration or graphically.
- Accurate with uniform doping profile.
- Independent of interface trap.

Substrate Doping Profile

- Based on the measurement of differential change in charge produced by a differential change in voltage.
- Accurate only when the measurement is done in depletion mode where the free carrier concentration depends strongly on dopant impurity concentration.

$$C_{si} = \frac{\varepsilon_{si}}{W} \Longrightarrow dw = \varepsilon_{si} \cdot d(\frac{1}{C_{si}})$$

$$\frac{1}{C_T} = \frac{1}{C_{ox}} + \frac{1}{C_{si}} \Longrightarrow d(\frac{1}{C_T}) = d(\frac{1}{C_{si}})$$

$$dQ_G = -q \cdot N(w) \cdot dw = -q \cdot N(w) \cdot \varepsilon_{si} \cdot d(\frac{1}{C_T})$$

N(w): Doping concentration as a function of depth

dw: Change of the depletion layer thickness

we also have: $dQ_G = C_T \cdot dV_G$

Therefore: $N(w) = -[q\varepsilon_{si} \cdot \frac{1}{C_T} \frac{d}{dV_G}(\frac{1}{C_T})]^{-1}$

$$N(w) = -\left[q\varepsilon_{si} \cdot \frac{1}{C_T} \frac{d}{dV_G} \left(\frac{1}{C_T} \right) \right]^{-1} = -2\left[q\varepsilon_{si} \cdot \frac{d}{dV_G} \left(\frac{1}{C_T^2} \right) \right]^{-1}$$

- Doping profile as a function of depth W can be obtained from the slope of $1/C_T^2$ as a function of V_G curve.
- Limitations of this method:
  - The above equation measures the majority carrier density in the vicinity of the depletion layer edge, rather than the dopant ion density itself.
  - The existence of interface charge gives errors.
  - The maximum depth to which doping profile can be measured is limited by the onset of inversion.

Flat Band Voltage and Oxide Thickness

V_{FB} from C-V

$$C_{si,FB} = \frac{\varepsilon_{si}}{L_D} \qquad L_D = \sqrt{\frac{\varepsilon_{si}}{qN_A\beta}} = \sqrt{\frac{2\varepsilon_{si}(\frac{kT}{2q})}{qN_A}} = \sqrt{\frac{\varepsilon_{si}(\frac{kT}{q})}{qN_A}}$$

$$\frac{1}{C_{FB}} = \frac{1}{C_{ox}} + \frac{1}{C_{si,FB}} \Longrightarrow$$ Get V_{FB} from C-V curve

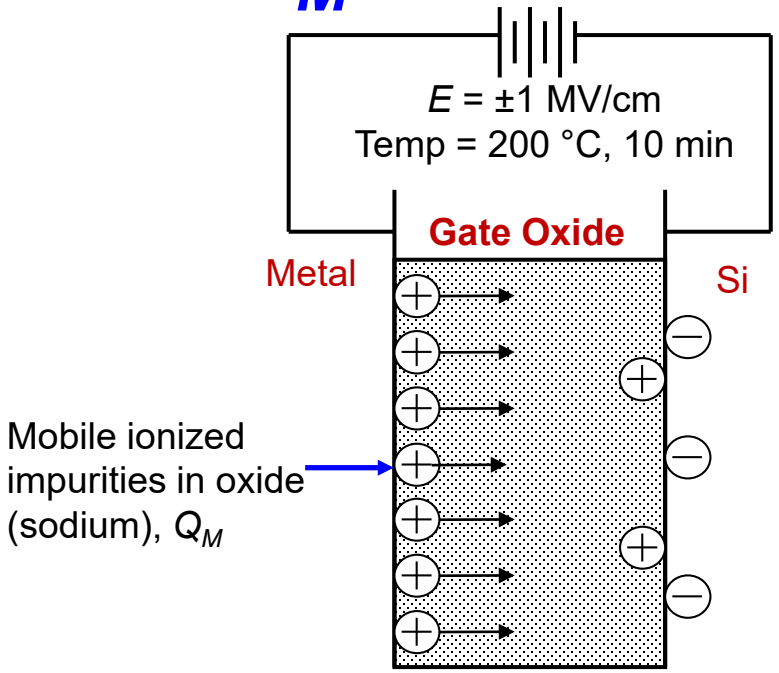
T_{ox} from C_{ox}

$$T_{ox} = \frac{\varepsilon_{ox}}{C_{ox}} \Longrightarrow$$ Get T_{ox} from C-V curve

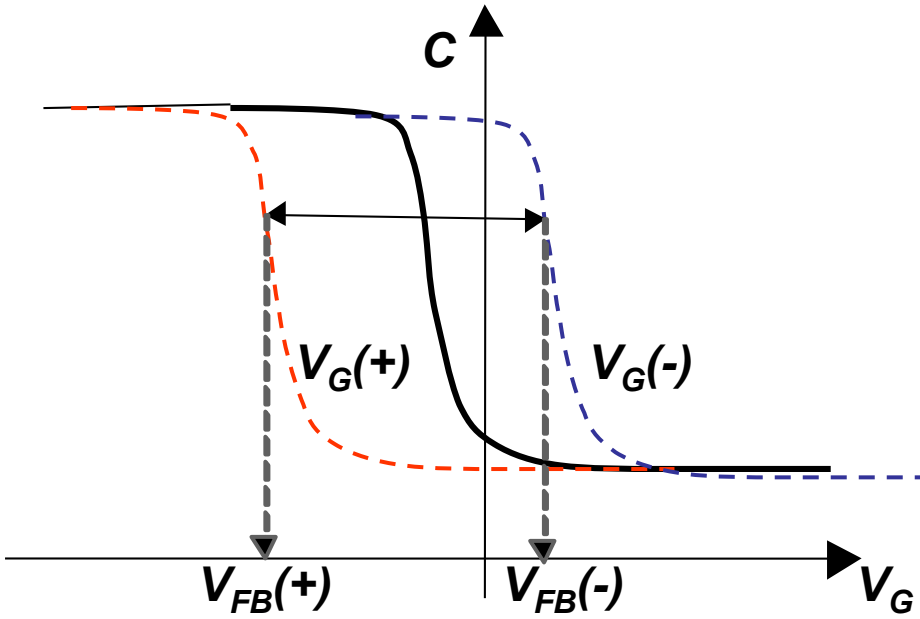
Measurement of Q_M

Bias-Temperature Stress

1. Measure initial *C-V*
2. Apply *E* > +1 MV/cm for 0~15 min @150 ~300 °C
3. Cool sample under bias
4. Measure *C-V* to get $V_{FB}(+)$
5. Apply *E* < -1 MV/cm for 10~15 min @150 ~300°C
6. Cool sample under bias
7. Measure *C-V* to get $V_{FB}(-)$



$$Q_M = \left| V_{FB}(-) - V_{FB}(+) \right| \cdot C_{ox}$$



Measurement of Q_f

1. From ideal *C-V*

- Assume all other charges Q_{it}, Q_M, Q_{ot} =0
- Calculate the ideal MOS *C-V* curve at a given T_{OX}, N_A, Φ_M

$$V_{FB} = \Phi_{MS} - \frac{Q_f}{C_{ox}}, \qquad \Phi_{MS} = \Phi_M - \Phi_S$$

$$\Phi_S = \chi + \frac{E_g}{2q} + \frac{kT}{q}\ln\left(\frac{N_A}{n_i}\right)$$

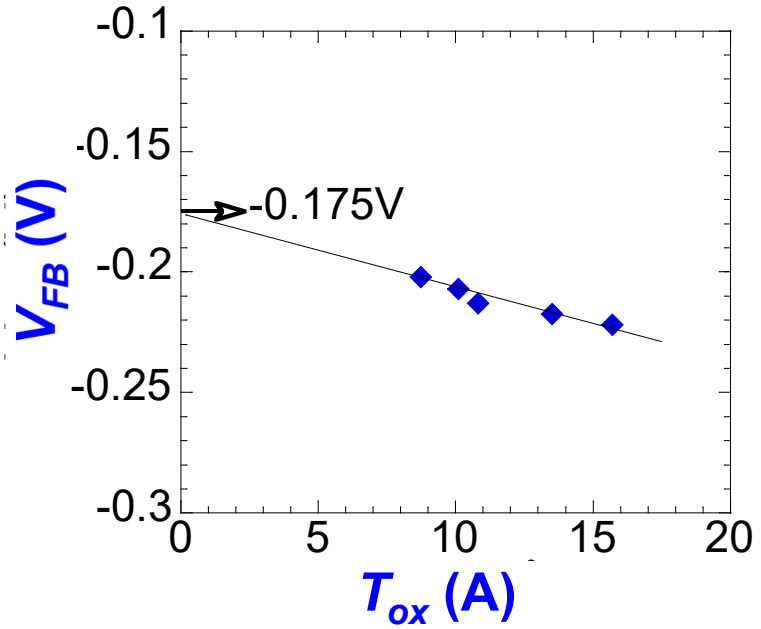
$$Q_f = C_{ox}\cdot\left(\Phi_{MS} - V_{FB}\right)$$

- Measure ΔV_{FB} between ideal *C-V* and measured *C-V*

$$Q_f = Q_{FB} = \Delta V_{FB}\cdot C_{ox}$$

2. From V_{FB} - T_{ox}

Q_f: Slope of V_{FB}-T_{ox} curve



Measurement of Q_{ot}

From hysteresis in *C-V*

- Q_{ot} will not distort the *C-V* curve but cause shifts on the voltage axis.
- Trapping and de-trapping processes cause hysteresis in the *C-V* curve.
- ΔV_{FB} between hysteresis curves gives the number of charges which are involved in trapping/de-trapping process.

